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A model-independent tool for evolutionary constrained multi-objective optimization under uncertainty

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ABSTRACT

An open-source tool has been developed to facilitate constrained single- and multi-objective optimization under uncertainty (CMOU) analyses. The tool uses the well-known PEST interface protocols to communicate with the underlying forward simulation, making it non-intrusive. The tool contains a built-in parallel run manager to make use of heterogeneous and distributed computing resources. Several popular and well-known evolutionary algorithms are implemented and can be combined with a range of approaches to represent uncertainty in model-derived constraint/objective values. These attributes serve to address the current barrier to adopt advanced CMOU analyses for a wide range of decision-support problems across the environmental modeling spectrum. We demonstrate the capabilities of the CMOU tool on a well-known analytical benchmark problem that we augmented to include uncertainty, as well as on a synthetic density-dependent coastal groundwater management benchmark problem. Both demonstrations highlight the importance of explicitly accounting for uncertainty to convey risk and reliability in pareto-optimal design.

1. Introduction

Increasingly, decision making for environmental resource management is informed by predictive models. Model-based decision support regarding unseen/future outcomes can be achieved in several ways, including with predictive scenario analyses, or through formal management optimization analyses, among others. Formal management optimization analyses are sometimes favored as they can offer a more "direct" means of predictive decision support (Ahlfeld and Mulligan, 2000; Maier et al., 2014; Gorelick and Zheng, 2015; Danapour et al., 2021). That is, while predictive scenarios are typically focused on simulating the outcome of "what if" conditions related to unseen (changes in) forcings, optimization analyses can extend the utility of predictive scenario analyses. For example, optimization analyses can be used to identify optimal resource management decisions that avoid unwanted outcomes while minimizing costs and/or maximizing resource allocation.

Optimization analyses can vary significantly in their formulation, including factors such as dimensionality (decision variables, objectives, and constraints), global versus local search methods, non-linear versus linear problem formulations, etc. In many decision-support settings, how best to conceptualize and formulate an optimization problem is subjective and requires investigation and testing. For example, a constrained management optimization problem may be better represented as a (constrained) multi-objective management optimization problem, where one or more inequality constraints are treated as an objective that can be traded off with other objectives (Deb, 2001; Nouiri et al., 2015; Siade et al., 2019; Paulinski et al., 2018). The interplay between constraints and objectives warrants efficient reformulating management optimization problems by treating one or more constraints as objectives and vice-versa—this leads to constrained multi-objective optimization (Fonseca and Fleming, 1998).

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When models are used to guide environmental resource decision making, the uncertainty associated with decision-relevant simulated outcomes (e.g. predictions, forecasts, constraints, objectives, etc) must be expressed. In many modeling settings, particularly subsurface settings, the inputs to the model are highly dimensional and highly uncertain, and there are typically few data available to condition uncertain model inputs, meaning that uncertainty in decision-relevant simulated outputs can be significant (Moore and Doherty, 2006; Herckenrath et al., 2011; Brakefield et al., 2015; Knowling et al., 2020a).

The need to quantify uncertainty in decision-relevant simulated outcomes, as well as to flexibly represent modeling analyses as constrained multi-objective optimization problems, gives rise to a complex, coupled analysis referred to herein as constrained multi-objective optimization under uncertainty (CMOU). Several previous works have investigated various aspects of CMOU in environmental modeling settings (see for example Singh and Minsker, 2008; Luo et al., 2016; Singh and Minsker, 2004; Mantoglou and Kourakos, 2007; Rezaei et al., 2020). These studies collectively (a) demonstrate the value that CMOU brings to decision support, (b) identify the broad range of settings that CMOU may be applied to, and (c) highlight the complexity and user requirements that implementing CMOU is likely to entail. Pertinent to this last point is that, to our knowledge, there exists no generally applicable tool that facilitates CMOU analyses in a non-intrusive (i.e., model-independent) and flexible way to support the wide range of environmental modeling settings that could benefit from CMOU.

To enable practitioners the ability to realize the benefits of CMOU analyses, we have developed pestpp-mou. pestpp-mou is an open-source tool that pairs several existing and well-known single-objective and multi-objective evolutionary algorithms with flexible approaches to represent uncertainties in model-derived outputs that serve as constraints and/or objectives in the optimization solution process. pestpp-mou has been designed to function in a non-intrusive and flexible way, using the well-known PEST model interface protocols, so that it can be easily applied to a wide range of CMOU problems that require different underlying simulation engines. To make use of parallel computation, pestpp-mou includes a built-in, fault-tolerant parallel run manager for heterogeneous and distributed computing systems. Although there are some already available software tools that implement certain forms of multi-objective optimization (see for example Kwakkel, 2017, Matott, 2005, and Hadka and Reed, 2013, among others), pestpp-mou is unique in that it implements constrained single-objective and multi-objective optimization under uncertainty in a non-intrusive and easy-to-deploy (e.g. statically linked, stand-alone) software tool.

2. Background and conceptual overview

As discussed in White et al., 2018, there is nuanced terminology associated with both uncertainty/risk analysis and management optimization analysis, particularly multi-objective optimization analysis. Herein we define the following terms:

- decision variable: an input to the model that is controlled by management action. The optimization process adjusts these quantities in seeking optimal solutions;
- simple constraint: an inequality quantity that is derived from combinations of decision variables. Because this quantity is derived purely from decision variable values, it is not subject to uncertainty;
- simple objective: a quantity that is derived from combinations of decision variables that is treated as an objective to minimize or maximize. Because this quantity is derived purely from decision variable values, it is not subject to uncertainty;
- model-derived constraint: an inequality quantity that is at least partially dependent on model outputs. Because this quantity depends on simulation results, it is subject to uncertainty in as much as it is sensitive to uncertain model inputs;
- model-derived objective: a quantity that is at least partially dependent on model outputs and that is treated as an objective to minimize or maximize. Because this quantity depends on simulation results, it is subject to uncertainty in as much as it is sensitive to uncertain model inputs;
- parameter: a quantity representing uncertain model inputs. Collectively, the uncertainty in parameters may induce uncertainty in model-derived constraints/objectives;
- individual: a single set of decision variable values;
- population: a collection of individuals, each of which has a value for each decision variable. The population is evolved through each generation of the CMOU solution algorithm towards an optimal set of solutions;
- generation: an iteration of the CMOU algorithm that consists of generating a new candidate population, evaluating the population by running each individual in the population through the model, and evaluating the resulting feasibility and fitness of each individual. Across multiple generations, the population evolves, emulating the processes of natural selection;
- feasible: an optimization solution (e.g. an individual) that satisfies all constraints;
- fitness: the metric used to rank individuals in a population. In the case of single-objective optimization, fitness is typically defined only by feasibility and objective function value. In multi-objective optimization, fitness includes feasibility, as well as pareto optimality/dominance and diversity/uniqueness elements.
- generator: an algorithmic process that emulates part of the evolutionary process by "generating" offspring from one or more parent individuals in the current population; and
- selector: an algorithmic process that emulated part of the evolutionary process by comparing all members in a population to determine "fitness", which can then be used by the generator to propagate desirable characteristics to the next generation.

2.1. Evolutionary meta-heuristic multi-objective optimization

Evolutionary algorithms are an important and exciting branch of research that have applications across the computational sciences (e.g. Michalewicz et al., 2003). In the most general sense, this class of search algorithms employs the Darwinian concept of "survival of the fittest" or natural selection, and uses characteristics and behaviors observed in nature to generate and evolve a population of individuals towards optimality. Herein, we only briefly review the concepts related to multi-objective optimization. Interested readers are referred to Maier et al., 2014; Siade et al., 2019 for more detailed information regarding evolutionary computation in the context of environmental resource management.

How optimality is defined is a critical aspect of any multi-objective evolutionary algorithm, and may include elements of feasibility, diversity, and pareto optimality. Pareto optimality is a central concept in multi-objective optimization. We define an individual i of a population as "pareto optimal" (or "pareto dominant") compared to individual j if for all objectives, i has an equal or better value compared to j **and** i has a strictly better value than j for at least one objective (e.g. Zitzler et al., 2003; Emmerich and Deutz, 2018). Conceptually, a solution is pareto optimal if for all objectives, no single objective can be improved further without compromising the value of another objective. The goal then of a multi-objective optimization algorithm is to find a population of pareto-optimal individuals. Collectively, a population of pareto-optimal individuals map the optimal trade-off between non-commensurate and competing objectives. This trade-off surface is referred to as the "pareto frontier".

2.2. Chance constraints and objectives

Because model-derived constraints/objectives may depend (at least partially) on model outputs, and because all models of environmental systems are subject to uncertainty, it follows that these model-derived quantities are also uncertain. The extent to which they are uncertain depends on many factors including the system that is being modeled, the model itself, and the available information (both prior estimates of the model inputs and historic state observations to be assimilated, etc), as well as many other factors. The focus of predictive uncertainty analysis is to quantify uncertainties in unmeasured, decision-relevant model outputs and possibly also to reduce these uncertainties through data assimilation (e.g. from prior to posterior uncertainty estimates).

Within CMOU analyses, the uncertainties in the model-derived constraints/objectives are the same uncertainties that would arise if the model-derived constraints/objectives were instead considered predictions within an uncertainty/data assimilation analysis. Herein, to account for the range of plausible constraint/objective values arising from model-input uncertainty within the (constrained) multi-objective optimization solution process, we employ the concept of "chance constraints" and/or "chance objectives". The "chance" in chance constraints/objectives represents the combination of model-derived constraint/objective uncertainty with a specified "risk" value to collapse the constraint/objective uncertainty to a single representative value. The value of risk expresses the level of aversion to an unwanted outcome. For example, a risk value of 0.95 expresses a 95% chance of avoiding an unwanted outcome—a risk-averse, high-reliability scenario—while a risk value of 0.05 expresses only a 5% chance of avoiding an unwanted outcome—a risk-tolerant, low-reliability scenario. Conceptually, the desired risk value is combined with the uncertainty in model-derived constraints/objectives by "shifting" the simulated values toward or away from the desired outcomes along the range of possible values. Interested readers are referred to (Wagner and Gorelick, 1987; White et al., 2018, 2020) and the references cited therein for more background on the concept of chance constraints and risk.

2.3. Coupling

In CMOU analyses, models are used to evaluate two fundamental relations: (a) the relation between decision variables and model-derived constraints/objectives (the decision variable relation), and (b) the relation between uncertain parameters and model-derived constraints/objectives (the parameter relation). If changes in the decision variable values influence the parameter relation, or changes in the parameters influence the decision variable relation, then this interaction should be considered within the CMOU analysis because it may affect the optimal, risk-based solution(s). Herein, we refer to the interaction between the uncertainty/risk analysis (which relies on the parameter relation) and the management optimization analysis (which relies on the decision variable relation) within the larger CMOU analysis as "coupling" (Lopez and Beck, 2012; Hamzehkolaei et al., 2016).

If no coupling exists, then the chance/risk estimates are independent of the decision variable values, meaning the chance/risk estimates can be used across multiple generations of the CMOU process and shared between individuals within a population—the uncertainty estimates calculated at one point in decision variable space are valid across decision variable space. However, if a coupling exists, then as the values of decision variables change across generations and/or across decision variable space, the relation between the uncertain parameters and model-derived constraint/objective values may need to be re-evaluated. For example, in designing a groundwater contaminant-transport treatment system, as the treatment rate and location of extraction and injection wells changes (as part of an optimal design search), the relation between uncertain properties (e.g., hydraulic conductivity, porosity) and the simulated concentration at compliance points (e.g. constraints/objectives) will be affected because the flow field induced by differing treatment designs may "activate" the influence of previously insensitive uncertain parameters (Knowling et al., 2020b).

Given the problem-specific nature and wide range of coupling possibilities, we recognize three difference approaches to represent uncertainty in model-derived constraints/objectives:

1. specified uncertainty approach ("specified");
2. First-order second moment (FOSM) approach ("FOSM-based");
3. stack-based approach ("stack-based").

These approaches imply different uncertainty conceptualizations, as well as different representations of the coupling between the uncertainty/risk analysis and the management optimization analysis within the larger CMOU analysis. Flexibility in how uncertainty and the corresponding chances and risk are represented in the CMOU analysis is a key consideration for undertaking CMOU analyses because each approach represents a trade-off between computational burden and rigorous representation of uncertainty in model-derived constraints/objectives. We note that each of the following approaches can be employed as either prior or posterior formulations, depending on the availability of state observations to condition parameters. It is expected that in practice, most applications of CMOU will use posterior uncertainty estimates, and that these posterior estimates will be derived from an "upstream" (e.g previously completed) data assimilation and uncertainty analysis.

2.3.1. "Specified" approach

The simplest and most computationally efficient uncertainty representation is the "specified" approach. As the name implies, this approach relies on directly specifying the uncertainty (e.g., standard deviation) of model-derived constraints/objectives with an assumed Gaussian probability distribution, with the mean of the distribution being the simulated value for the constraint/objective. By specifying constraint/objective uncertainties, no additional model evaluations are required in the CMOU solution process. However, this approach is only appropriate if there is no expected coupling between the uncertainty/risk analysis and the management optimization analysis. The "specified" standard deviations would typically be obtained from a previous uncertainty/data assimilation analysis; parameters are not explicitly used in this formulation.

2.3.2. FOSM-based approach

A more robust approach for representing constraint/objective uncertainty is with FOSM analyses (see e.g., White et al., 2018, White et al., 2016, Doherty, 2015, Menke, 1989, Goldstein and Wooff, 2007). Like the specified approach, the FOSM representation also implies the uncertainty in the model-derived constraints/objectives can be approximated by a Gaussian distribution, and the mean values for these implied Gaussian distributions are the simulated values. However, the FOSM approach involves calculation of these uncertainties via (conditional) covariance propagation from (uncertain) parameters to the model-derived constraints/objectives. In this way, users provide the uncertainties of the parameters via a statistical parameter distribution, and the parameter uncertainties are propagated to the model-derived constraints/objectives "on-the-fly" during the CMOU analysis. The FOSM-approach requires a Jacobian matrix of sensitivities of model-derived constraints/objectives to the uncertain parameters. Furthermore, the FOSM-based approach allows for some accommodation of coupling between the risk/chance analysis and the management optimization analysis since the Jacobian matrix (and the corresponding FOSM-based constraint/objective uncertainties) can be updated as the CMOU analysis moves through generations and the region of focus in decision-variable space changes. However, the cost of repeated evaluation of the Jacobian matrix, especially when the number of parameters is large, is computationally expensive.

2.3.3. Stack-based approach

The most robust approach for representing uncertainties in the model-derived constraints/objectives is with "stacks" (Bayer et al., 2008). In this representation, an ensemble or "stack" of parameter realizations is used to represent model input uncertainty, essentially using Monte Carlo approach to account for these uncertainties. Compared to the previous two approaches, the stack-based approach makes no assumption about the statistical distribution of the model-derived constraint/objective uncertainties; that is, the stack-based approach is a non-parametric representation of constraint/objective uncertainties. The stack-based approach uses the model-derived constraint/objective cumulative distribution function arising from evaluating the parameter stack at a given point in decision-variable space. Once the stack of parameter realizations is evaluated, each model-derived constraint/objective has a "stack" of possible values, this set of possible values represents the range of values that model-derived constraints/objectives may take.

3. Methods and implementation

pestpp-mou has been designed to implement a wide range of CMOU analyses to support environmental resource decision making. It is a statically-linked C++ program that uses the well-known PEST model interface protocols (Doherty, 2015) to communicate with the underlying simulation in a non-intrusive and flexible way. Given that CMOU analyses require many model forward evaluations and, inevitably, some of these evaluations may lead to forward model convergence failure or other model instability issues, pestpp-mou includes a built-in, fault-tolerant parallel run manager. The run manager uses TCP/IP to communicate, a lower-level and more common network protocol than the message passing interface (Clarke et al., 1994), which allows pestpp-mou to be deployed on a wide range of heterogeneous computing clusters, including ad hoc networks and in cloud settings (see Fienen and Hunt, 2015 for more information on this important feature). The non-intrusive design and flexible parallel run management capabilities of pestpp-mou are key contributions of this work, as these aspects enable CMOU analyses for many environmental management settings.

3.1. Generators and environmental selectors

The field of evolutionary multi-objective optimization is rapidly evolving and several tools that allow practitioners and researchers to explore capabilities of these algorithms already exist (e.g. Durillo and Nebro, 2011; Kwakkel, 2017; Hadka and Reed, 2013). Rather than develop yet another entry in this already crowded field, we have focused instead on developing a production-level tool that implements several commonly used and previously vetted evolutionary algorithms with flexible and generic uncertainty/risk evaluation in non-intrusive way with built-in, fault-tolerant parallel run management. As presently implemented, pestpp-mou offers several algorithmic choices for evolutionary generation and selection, all of which have displayed strong multi-objective optimization performance in the literature (e.g. Reed and Hadka, 2014; Durillo and Nebro, 2011). pestpp-mou purposely separates the generator and selector, essentially breaking apart the two primary evolutionary mechanisms so that users have greater flexibility on how to combine these two algorithmic elements. To be clear, the generator produces new individuals from the existing population. These new individuals are then evaluated through the model and their fitness is calculated by the selector.

Choices for evolutionary generators include the following:

- differential evolution (DE) (Storn and Price, 1997) (including a self-adaptive variant (Georgioudakis and Plevris, 2020) whereby the crossover and mutation rates are treated as decision variables)
- particle swarm optimization (PSO) (Kennedy and Eberhart, 1995; Siade et al., 2019),
- simulated binary crossover (SBX) (Deb and Agrawal, 1995), and
- highly-disruptive polynomial mutation (PLM) (Hamdan, 2012).

pestpp-mou users can also employ combinations of these generators. This is achieved by simply specifying more than one generator. We have also included the reducing-set concurrent simplex (Lewis et al., 2006) (SIMPLEX), a version of the Nelder-Mead simplex (Olsson and Nelson, 1975; Lagarias et al., 1998) with increased parallelism, as a gradient-based generator in pestpp-mou.

As presently implemented, pestpp-mou uses the constrained NSGA-II environmental selector proposed in Deb et al., 2002. The code also implements the unconstrained SPEA-2 environmental selector described in Zitzler et al., 2001.

Just as with the generators, we recognize that much past and current research has been and is being devoted to studying constraint handling in multi-objective evolutionary optimization (e.g. Kramer, 2010; Huang and Wang, 2021; Mezura-Montes, 2009; Fonseca and Fleming, 1998; Aguirre et al., 2004; Liu et al., 2021). As such, we have relied on previously published approaches that have demonstrated success in constraint handling in applied environmental modeling studies to guide the current implementation in pestpp-mou (e.g. Majedi et al., 2021; Fan et al., 2020; Mirzaie-Nodoushan et al., 2017; Naghdi et al., 2021; Raei et al., 2017).

We note that if a single objective is specified, pestpp-mou seamlessly functions as a (constrained) single-objective evolutionary optimization tool, and the same uncertainty/risk functionality is still available for the single objective and any constraints that are included (provided these quantities are model-derived). This seamless flexibility allows users to efficiently move between constraints and objectives to evaluate different optimization problem formulations.

3.2. Chance constraints/objectives

pestpp-mou offers the three previously described approaches to represent model-derived constraint/objective uncertainty: specified, FOSM-based, or stack-based. pestpp-mou can automatically draw the parameter stack for the listed parameters using either basic initialization information (e.g. initial values and parameter bounds) by assuming the parameters are statistically independent.

In the case that pestpp-mou is being used "downstream" from a data assimilation/uncertainty analysis, users can provide a posterior parameter covariance matrix (e.g. Doherty, 2015; White et al., 2016; White, 2018) for stack generation and/or for FOSM uncertainty representation; users can also supply a previously generated stack, such as a parameter ensemble derived from a posterior uncertainty analysis. pestpp-mou also automates the process of populating and evaluating the Jacobian matrix needed in the FOSM-based approach. Through shared file types related to the chance process, pestpp-mou provides a seamless transition from uncertainty/data assimilation analyses undertaken with other tools in the PEST++ and PEST suites to CMOU analyses. We note that the entire risk/uncertainty process is not required to apply pestpp-mou and is only activated by specifying a risk value not equal to 0.5.

pestpp-mou includes two options for where in decision-variable space to evaluate stack-based chances through the "chance points" option, namely, at a single representative point ("single")—a computationally cheap but less rigorous option—or to evaluate the stack at all individuals in a population ("all"), a computationally expensive but more rigorous option.

Following the concepts presented in (Deb et al., 2007), pestpp-mou also provides the functionality to treat "risk" as an objective to be maximized (i.e., driven towards a risk-averse, high-reliability stance). In this way, if decision makers are unsure what risk stance is appropriate *a priori*, or users are unsure what maximum value of risk may lead to infeasibility, then specifying "risk" as a formal objective alleviates the requirement to specify a single risk value. Note, however, this option

will likely lead to an increased computational burden as additional generations will likely be required to accommodate an additional objective.

3.3. Coupling approaches

How and when the chance/risk estimates should be updated in the course of a CMOU analysis is likely to be problem specific. Given this, pestpp-mou strives to provide users with flexibility to accommodate a range of coupling approaches. Users can choose to "re-use" or update the risk/uncertainty estimates for a given generation flexibly with pestpp-mou to lower the computational burden of applying pestpp-mou. For the FOSM-based approach, only the "second moment" is used, so this implies constraint/objective uncertainty estimates can be broadcast across the population by simply applying the FOSM-estimated uncertainties in model-derived constraints/objectives to each individual's simulated values for these quantities. To update the FOSM estimates, the Jacobian matrix must be re-evaluated, which, when requested, is completed using the decision variable values nearest the current-generation mean decision variable values in a euclidean-distance sense.

In the case of a strong coupling within the CMOU analysis using stack-based uncertainty representation, the stack can be evaluated at each individual in the population, or, if a weaker coupling is present, the stack can be evaluated at a single representative individual (point in decision variable space) and broadcast to the remaining individuals, as an option to reduce the computational burden. The former requires many repeated stack evaluations, with a complete stack evaluation required at each individual in the current population (number of model evaluations equals the number of realizations in the stack times the number of individuals in the population), while the latter requires a single stack evaluation (the number of model evaluations equals the number of realizations in the stack). As such, the implementation of flexible coupling schemes with the stack-based approach requires additional consideration.

To re-use stack-based uncertainty/risk estimates across multiple generations, pestpp-mou maps each current individual solution to the nearest stack-evaluation point in decision-variable space in a minimum-Euclidean-distance sense. The nearest stack results are then translated to each corresponding current population individual by differencing the mean value of the stack results with the simulation results of the current individual. In this way, the available stack results nearest each individual are used to represent constraint/objective uncertainty at that individual without requiring a re-evaluation of the stack, again reducing the computational burden.

4. Example applications

We demonstrate the functionality of pestpp-mou through two example applications. The first is a well-known analytical constrained multi-objective benchmark problem that we augmented in order to transform this problem into chance-constrained/chance-objective problem. This problem is used to benchmark pestpp-mou, facilitate comparisons between the performance of different evolutionary generators, as well as demonstrate how chances and risk affect the shape and orientation of the constrained pareto frontier in objective space.

The second example application is a variant of a popular saltwater intrusion benchmark problem, developed for the purpose of designing a hypothetical artificial recharge basin to maintain viability of coastal groundwater extraction wells. This problem is used to demonstrate decision-support CMOU capabilities that pestpp-mou enables.

4.1. Benchmark problem

The analytical constrained multi-objective problem that we used for benchmarking is known as CONSTR (Deb, 2001; Deb et al., 2002). The CONSTR problem is formally defined as follows:

minimize: $f_1(\mathbf{x}) = x_1 + \theta_1, f_2(\mathbf{x}) = \frac{1+x_2}{x_1} + \theta_2$,
subject to: $x_1 \in [0.1, 1.0]$, $x_2 \in [0, 5]$,
and: $g_1(x) = x_2 + 9x_1 \geq 6$, $g_2(x) = -x_2 + 9x_1 \geq 1$.

Where θ_1 and θ_2 are additive uncertain parameters. As shown on Fig. 1, the pareto frontier between f_1 and f_2 (red dashed) is truncated by the edge of the feasible region (shaded region). As a result, the optimal pareto frontier is partially defined by the trade off between f_1 and f_2 and partially defined by constraints.

We modified the CONSTR problem to include two uncertain parameters (θ_1 and θ_2) whose action is additive for each of the two objectives (f_1 and f_2) respectively. This introduces uncertainty into the calculated f_1 and f_2 objective values. To be clear, this additive uncertainty results in no interaction between the risk/uncertainty analysis and the optimization analysis, so we use this uncoupled problem to verify the behavior of the various uncertainty representations and coupling strategies. For this problem, we conducted several experiments:

- specified risk values of 0.05, 0.5 and 0.95 using a "specified" uncertainty representation, 100-individual population and 50 generations;
- risk-as-an-objective using a "specified" uncertainty representation, 100-individual population and 150 generations;
- risk-as-an-objective using a stack-based uncertainty representation, evaluating the stack at a single chance point and reusing this single stack across all generations, 100-individual population and 150 generations; and
- risk-as-an-objective using a stack-based uncertainty representation, evaluating the stack at all individuals in the initial population and reusing these stack evaluations across all generations, 100-individual population and 150 generations.

The specified-risk analyses represent a risk-tolerant (0.05), risk-neutral (0.5), and risk-averse (0.95) stance, respectively. Treating risk as an objective will likely increase the complexity of the CMOU analyses, so the number of generations was increased to 150, making the results of the risk-as-an-objective analyses comparable to the combined results of the three specified-risk analyses with respect to computational budget.

4.1.1. Results

The application of pestpp-mou to the modified CONSTR benchmark verifies certain aspects of the CMOU solution process encoded in pestpp-mou and also demonstrates some of the more advanced capabilities of pestpp-mou. Fig. 1 A and B show that, for the risk neutral case (risk value of 0.5), the algorithm encoded in pestpp-mou finds the correct pareto frontier with either the DE or PSO generator. Furthermore, the risk averse and risk tolerance solutions (Fig. 1 A through F) are located in objective function space as expected. That is, they are shifted in the appropriate direction with respect to the risk-neutral pareto frontier: the risk-tolerant solutions occupying a larger feasibility region, while the risk-averse solutions occupy a smaller feasible region. Even in this analytical benchmark problem with simple additive objective uncertainty, the importance of accounting for uncertainty, and, ultimately, risk, is demonstrated by the change in pareto-optimal solutions as a function of risk.

The DE generator appears to have found more pareto optimal solutions than the PSO generator for the specified risk analyses; the DE pareto solution set point are more dense. However, we caution readers not to draw general conclusions regarding the relative performance of DE and PSO from these analyses because it is well known that there is no single algorithm (e.g., generator, selector, etc.) that is optimal for all possible optimization problems.

The use of the risk-as-an-objective option available in pestpp-mou is shown to perform as expected. Specifically, the results of the risk-as-an-objective approach (Fig. 1 C, D, E, F) appear to correspond appropriately

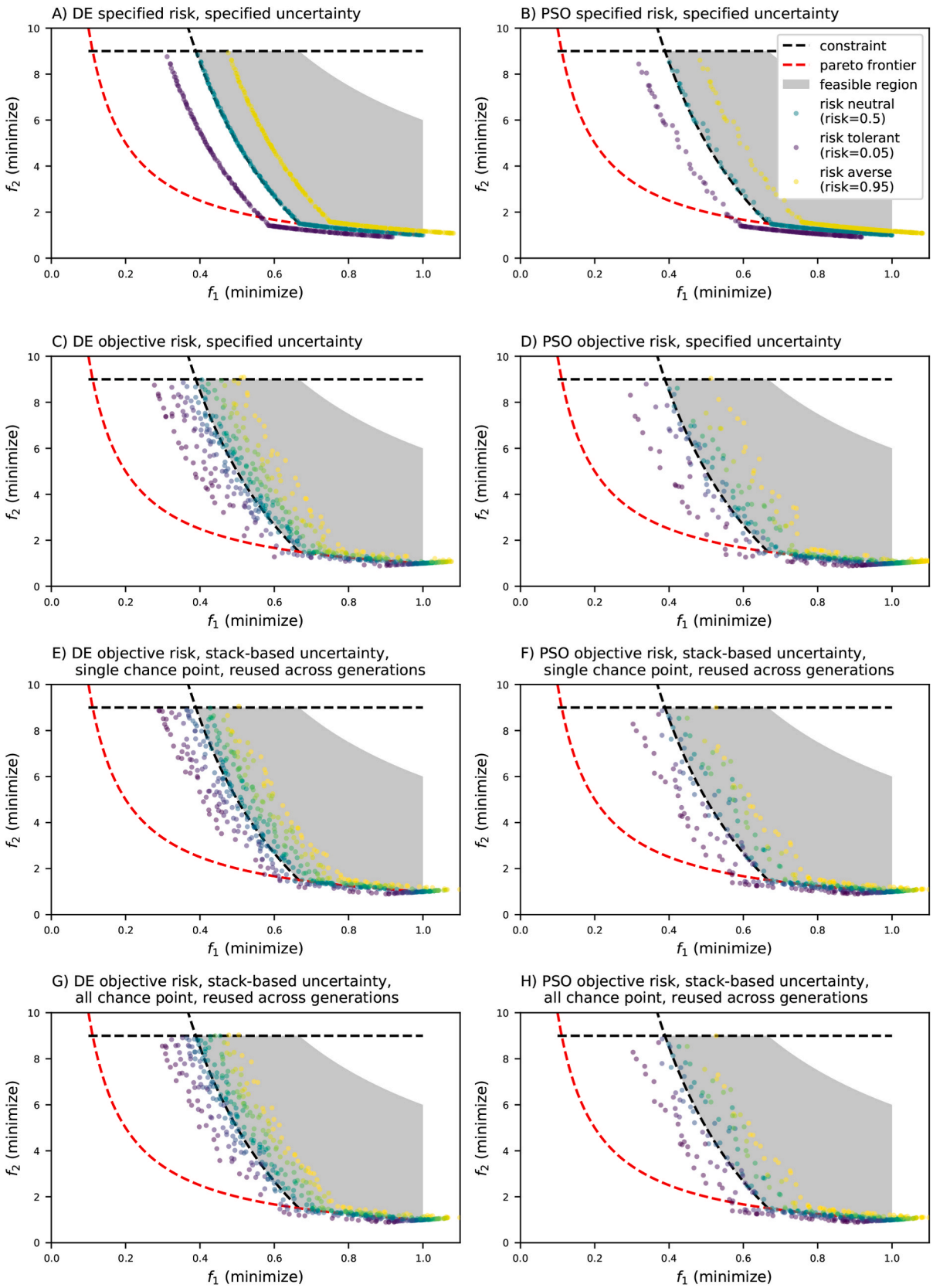


Fig. 1. Application of pestpp-mou's DE generator (A,C,E,G) and PSO generator (B,D,F,H) to the modified CONSTR benchmark problem for different approaches to accommodating risk and coupling strategies. The pareto frontier (red dashed line) is truncated by the feasible region (shaded), such that optimal solutions occupy a portion of the pareto frontier and a portion of a constraint. The risk-tolerant solutions project into the infeasible region (dark blue/purple solutions) and beyond the pareto frontier, while the risk-averse solutions contract away from the constraints and the pareto frontier (light yellow solutions). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

to explicit risk averse and risk tolerant pareto frontiers (Fig. 1 A and B). Additionally, given that this problem was designed to have no coupling between the risk/uncertainty analysis and the optimization analysis, we expect the outcome of varying the uncertainty representation and coupling strategies to yield similar results, verifying certain aspects of pestpp-mou functionality. For example, the results of specified-uncertainty approach (Fig. 1 C and D) appear very similar to the stack-based results (Fig. 1 E and F).

4.2. Henry problem

To demonstrate the risk-based decision support capabilities of CMOU via pestpp-mou in the context of coastal groundwater management, we evaluate a modified form of the cross-section Henry saltwater intrusion problem (Henry, 1964). The Henry problem is patterned after the coastal aquifer conditions in south Florida; while its spatial and temporal dimensions are representative of a laboratory scale experiment, the Henry problem is well-known (e.g. Herckenrath et al., 2011; Simpson and Clement, 2004; Zidane et al., 2012; Holzbecher, 2016; Croucher and O'Sullivan, 1995) and provides a computationally efficient test problem for density-dependent groundwater flow that captures many relevant aspects of coastal saltwater intrusion phenomena. The density-dependent characteristics of the Henry problem, combined with non-linear relations between model inputs and model outputs makes the Henry problem an illustrative benchmark for evaluating the CMOU algorithm encoded in pestpp-mou.

Starting with the Henry model design available with MODFLOW-6 (Hughes et al., 2017; Langevin et al., 2017) (see the MODFLOW-6 software release (Langevin et al., 2021) for details), we endeavored to design an artificial recharge basin to mitigate landward migration of the fresh-water/saltwater interface under increased groundwater extraction for water supply. The model used herein has the following details:

- 160 columns and 40 layers of 0.025 m spacing; columns are numbered starting farthest from the coastal boundary condition (Fig. 2). Note the water table is at or near 1.0 m in elevation.
- three vertically aligned groundwater extraction wells, located at column 60 in layers 10, 20, and 30 (Fig. 2);
- upgradient specified inflow boundary condition in column 1 and all layers (0.071 275 $\frac{m^3}{d}$ per layer) (e.g. Simpson and Clement, 2004) with a concentration of 0.0 $\frac{g}{l}$ salinity; and
- general-head boundary condition specified in column 160 and all layers to represent a coastal boundary with a stage of 1.0 m and concentration of 35.0 $\frac{g}{l}$ salinity.

Three distinct periods of time were simulated:

- pre-development period of 2.5 days to establish a stable freshwater-saltwater interface position without any groundwater extraction;
- historical groundwater use period of 0.3 day, which induces landward movement of the freshwater-saltwater interface towards the groundwater extraction wells (Fig. 2); and
- a predictive period of 2.5 days where the artificial recharge basin is in operation while groundwater extraction continues (or increases, depending on the selected decision variable values).

Within this simulation, we want to design a single artificial recharge basin, located on the surface (layer 1) between the coastal boundary and the upgradient inflow boundary to help maintain viability of the extraction wells to produce potable water (e.g. salinity less than or equal to 0.5 $\frac{g}{l}$).

The following decision variables subject to optimization were defined:

- location of the left (upgradient) edge of the recharge basin (column number from the left edge of the model domain) (minimum = 1, maximum = 140). Note the width of the recharge basin was fixed at 10 model cells wide;
- the rate of artificial recharge applied to the basin ($\frac{m}{d}$) (minimum = 0.001, maximum = 12.0);
- the concentration of recharge applied to the basin ($\frac{g}{l}$) (minimum = 0.5, maximum = 17.0); and
- the extraction rate of each of the three groundwater extraction wells ($\frac{m^3}{d}$) (minimum = 0.0, maximum = 3.0).
- The objectives of this paired recharge basin-groundwater extraction design CMOU analysis are:

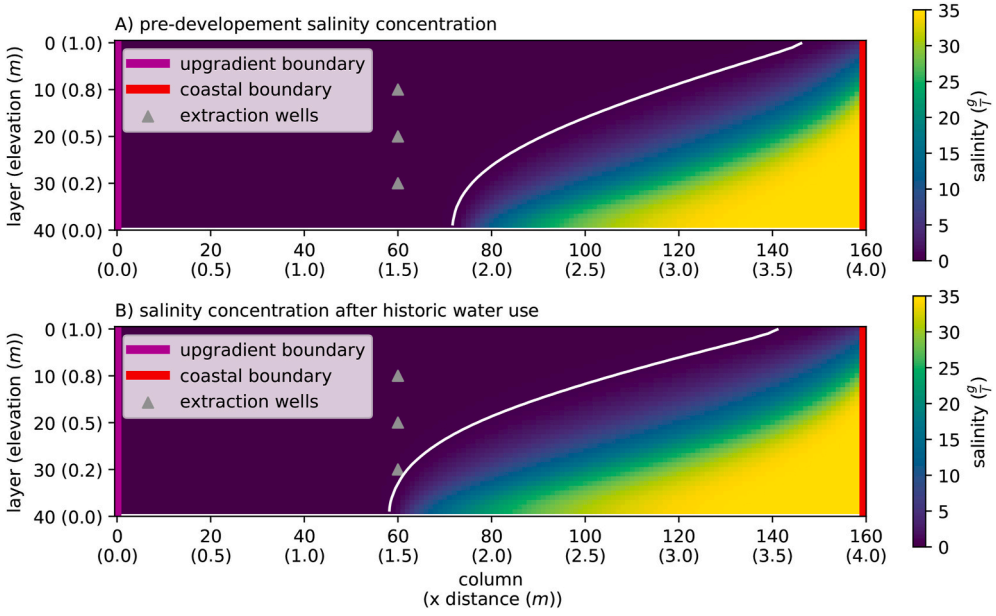


Fig. 2. The modified Henry model domain. The distribution of salinity is shown for pre-development conditions (A) and after historic groundwater extraction (B). The white line on both plots marks the 0.5 $\frac{g}{l}$ concentration (treated as the upper limit of salinity for potable purposes).

- maximize the artificial recharge basin distance from the upgradient inflow (prefer a more coastal location for the artificial recharge basin);
- minimize the artificial recharge rate applied to the basin (prefer to apply less artificial recharge);
- maximize the concentration of artificial recharge applied to the basin (prefer to apply low-quality water); and
- maximize the combined groundwater extracted from the three extraction wells.

The problem is subject to the following constraints:

- the simulated combined (flux-weighted mean) salinity of the three extraction wells is less than 0.5 $\frac{g}{l}$ (e.g. potable water);
- the simulated maximum water level within the recharge basin is less then 1.025 m (0.025 m allowable increase in water table elevation resulting from the recharge basin); and
- the combined flux from the three extraction wells is greater than 0.7 $\frac{m^3}{d}$ (the combined historical extraction rate).

Uncertainty arises in this optimization problem formulation through the concentration and groundwater level constraints. Because the model is used to evaluate the concentration at the extraction wells and groundwater level within the artificial recharge basin for a given set of decision variables, these simulated concentrations and groundwater levels are uncertain inasmuch as the uncertain model inputs influence these simulated outputs. We defined the following parameters to represent model input uncertainty:

- hydraulic conductivity in every active model cell ($\frac{m}{d}$) (mean = 864, minimum = 86.4, maximum = 8640.0);
- porosity in every active model cell (dimensionless) (mean = 0.35, minimum = 0.3, maximum = 0.4); and
- upgradient inflow (column 1, every layer $\frac{m^3}{d}$) (mean = 0.071 275, minimum = 0.064 147 5, maximum = 0.784 025).

A prior-uncertainty stance was used in this demonstration. Correlation in the spatially distributed parameter types was defined using an exponential variogram with a range of 0.1 m and sill proportional to the parameter bound range. A stack of 100 parameter realizations was used to represent parameter uncertainty in the pestpp-mou analyses. The coupling between the risk/chance estimates and the optimization problem was defined such that the stack was evaluated at every individual of the first generation and then reused for all subsequent generations.

Two pestpp-mou analyses were completed with the modified Henry model artificial recharge design problem. First, a risk-neutral analysis was completed to better understand possible (feasible) artificial recharge basin designs and trade-offs assuming the model-derived constraints are subject to no uncertainty; the risk-neutral analysis used 100 individuals and was evolved across 100 generations. A second pestpp-mou analysis was completed with risk-as-an-objective using 100 individuals and 500 generations; the 100-realization parameter stack was used to represent model input uncertainty in this analysis.

The Henry-based optimization analysis used the DE generator and the constrained NSGA-II environmental selector. Note the optimization problem occurs solely on the predictive period. However, to include the stack-based chances and to ensure these chance estimates are free from temporal artifacts, all simulation periods were evaluated for each model run.

The pestpp-mou input datasets for this analysis were generated programmatically using pyEMU (White et al., 2016) using the PstFrom automation process (White et al., 2021).

4.2.1. Results

Several interesting insights are gained by inspecting risk-neutral outcomes (Fig. 3). First, we note that through the use of an artificial recharge basin combined with formal optimization, substantially larger quantities of potable water can be produced compared to the more naive production strategy used during the historical period —a strategy that led to landward migration of the freshwater-saltwater interface; only solutions that exceed the naive production are shown on Fig. 3.

Among the most interesting insights gained from the risk-neutral analysis is the pareto-optimal relation between combined groundwater extraction rate and artificial recharge rate (Fig. 3 M and P). There is an apparent bimodal relation between these two important objectives, seen as two distinct lobes of optimal solutions. One mode involves an approximately low ratio of recharge to extraction and uses relatively low-salinity water, whereas the other mode involves a much higher ratio of recharge to extraction and uses high-salinity water (Fig. 3 A). The physical location aspect of these modes is key to understanding. The low-ratio, high-quality water mode includes artificial recharge basins located landward of the extraction wells (Fig. 3 A), essentially augmenting fresh upgradient inflows with high-quality (low-salinity) water to mitigate landward movement of the fresh-water/saltwater interface (Fig. 3 C, F). The high-ratio, low-quality water mode includes recharge basins that are located seaward of the extraction wells, where large volumes (relative to the combined extraction rate) of low-quality (high-salinity) water are injected to vertically depress the freshwater-saltwater interface and establish a hydraulic divide, preventing high-salinity seawater from moving landward in all but the deepest parts of the simulated aquifer—these solutions generally involve shifting extraction to the shallowest extraction well (located in layer 10). This dual-mode behavior highlights the capabilities of constrained evolutionary multi-objective optimization to identify less-intuitive, but feasible and (pareto-)optimal solutions.

Analyzing risk and chance for the Henry design problem brings substantially more context and insight. First, by allowing risk-tolerant solutions (risk less than 0.5), we see a wider range of possible extraction rates (for example, Fig. 4 K and W versus Fig. 3 E and N). We also now see that the low-ratio, high-quality-water design mode is generally lower in reliability than the high-ratio, low-quality-water design mode (Fig. 4 U and Y). This is likely because the low-ratio design is sensitive to both the subsurface properties of the system (e.g. hydraulic conductivity and porosity) and to upgradient inflow quantities, all of which were specified as uncertain inputs to the model. Not surprisingly, we also see that optimal solutions with very high combined groundwater extractions rates are also associated with lower reliability values (Fig. 4 F and V)). These solutions rely on a delicate balance between extraction rate and recharge rate, which can be disrupted by the uncertain parameters, allowing for landward migration of the freshwater-saltwater interface. Ultimately, the most reliable solutions use a seaward recharge basin (Fig. 4 D and L) with high ratios of recharge to extraction rate (Fig. 4 U and Y) and relatively low extraction rate (Fig. 4 F and V) (note only solutions that are three times or greater than the historical extraction rate are shown); the salinity of the recharging water can increase with increasing seaward distance of the basin (Fig. 4 K and W). These important insights, which are not available with risk-neutral analyses, highlight the importance of the CMOU workflow that pestpp-mou enables.

5. Discussion and conclusions

We have presented the CMOU workflow-enabling tool pestpp-mou. pestpp-mou implements several popular evolutionary algorithms and pairs these algorithms with several options for representing uncertainties in model-derived constraints/objectives. We have designed pestpp-mou to work in a “downstream” sense of the existing uncertainty and data assimilation tools in the PEST and PEST++ suites so that practitioners have more options to seamlessly transition from more

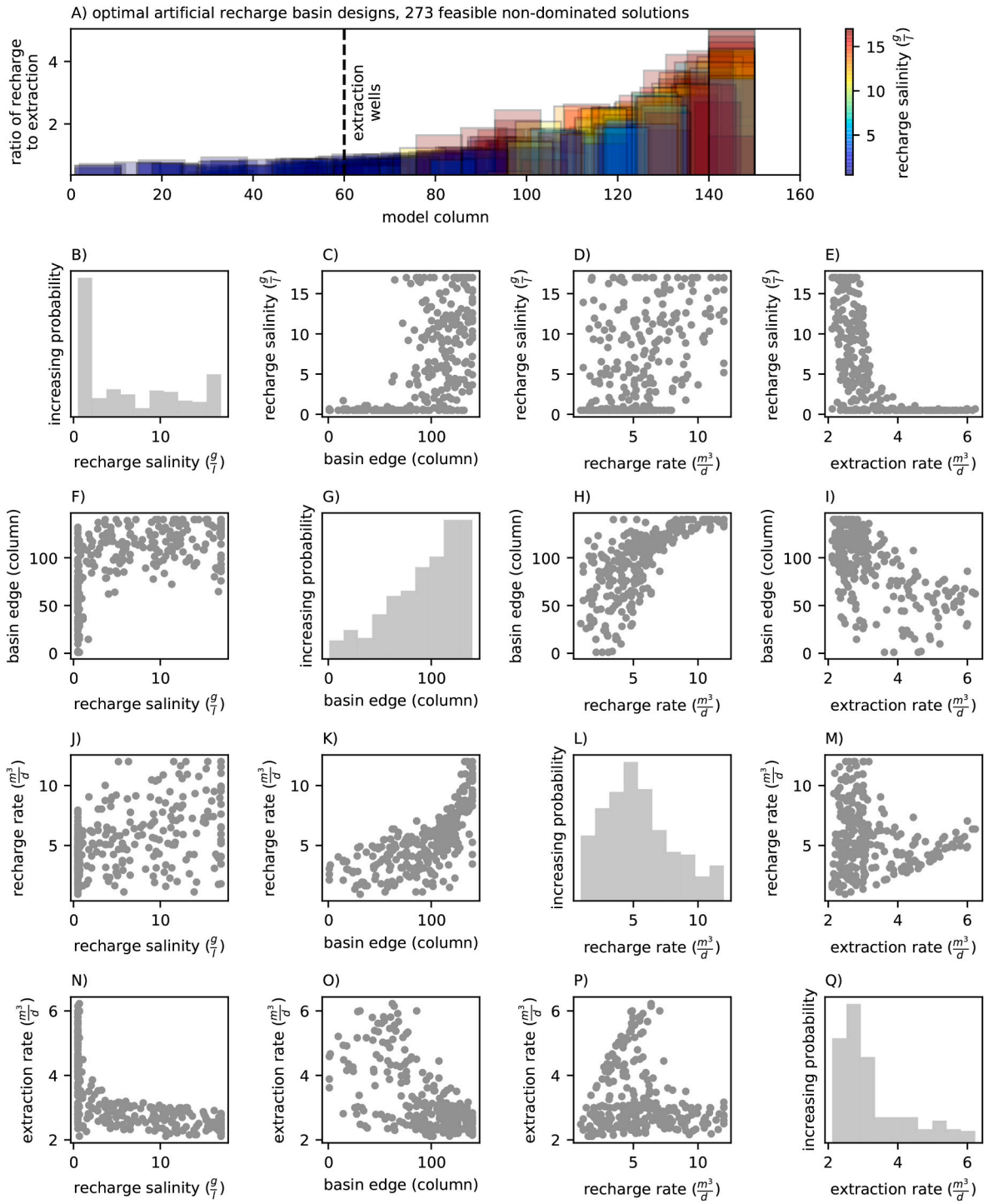


Fig. 3. Summary of risk-neutral CMOU analyses with the modified Henry model. The diagonal panes present the objective histograms of the pareto optimal solutions, while the off-diagonal panes are the cross-plots between given objectives; each marker on the cross plots represents a pareto optimal solution. Only solutions with combined extraction greater than 2.1 $\frac{m^3}{d}$ (three times the historical rate) are shown. A) displays the spatial characteristics of each pareto-optimal artificial recharge basin. Two optimal design modes were identified (M,P).

traditional uncertainty analyses and data assimilation to CMOU analyses, ultimately leading to better model-based decision support. This aspect of the pestpp-mou design, paired with the wide-spread use of the PEST model interface and the fault-tolerant parallel run manager embedded within pestpp-mou, makes this tool unique among others. The ability to formulate and evaluate complex CMOU problems in flexible and general ways allows pestpp-mou to be deployed to a wide range of decision-support analyses across the environmental modeling spectrum.

The choice of how to best couple the risk/chance analyses with the (constrained and/or multi-objective) management optimization analyses is a complex subject matter and will almost certainly be application specific. While practitioners may be tempted to err on the side of caution and re-evaluate the interaction between parameter and decision variable relations at every individual for every generation, the computational burden of this formulation is very high. Practitioners are instead

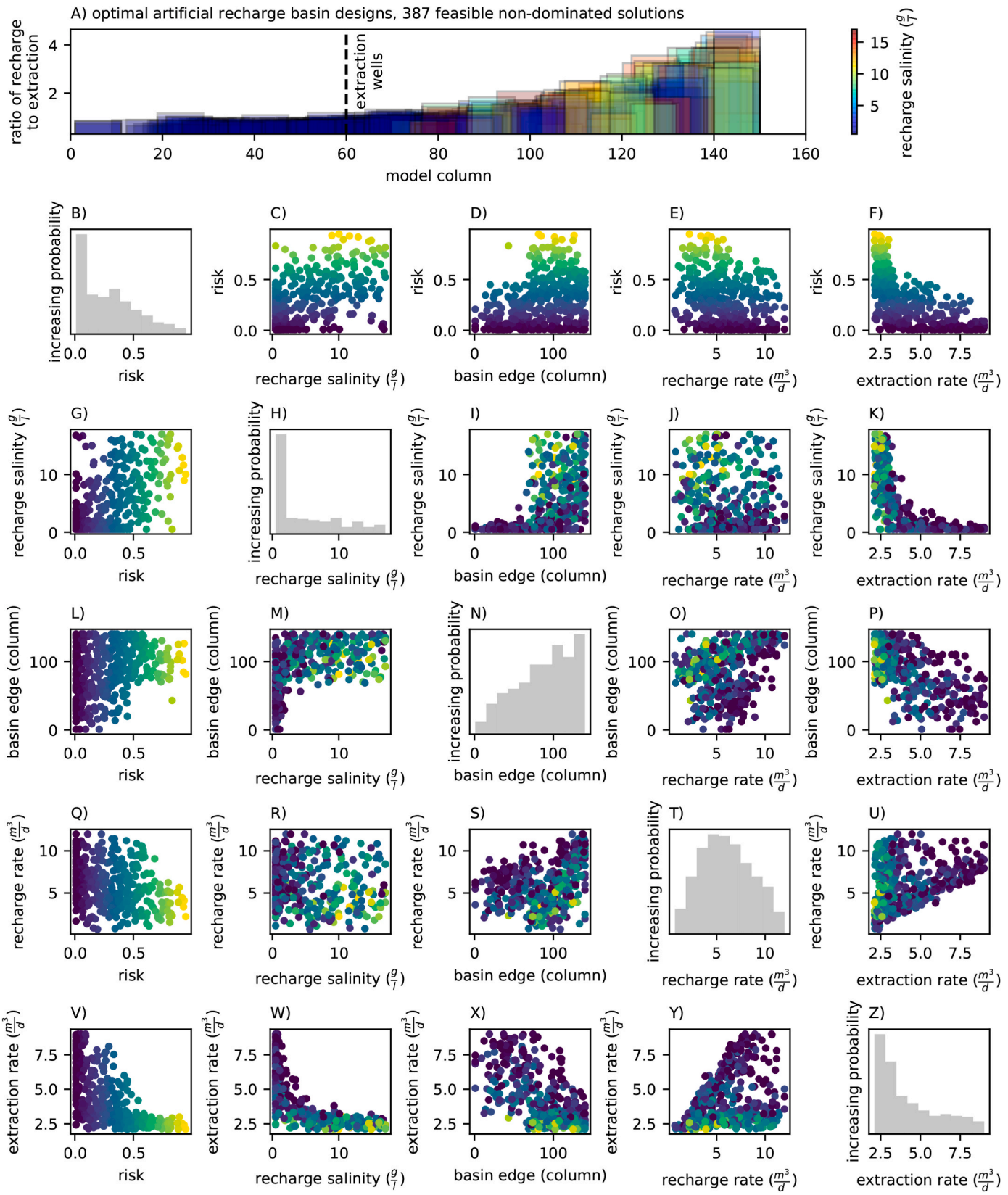


Fig. 4. Summary of risk-based CMOU analyses with the modified Henry model. The diagonal panes present the objective histograms of the pareto optimal solutions, while the off-diagonal panes are the cross-plots between given objectives; each marker on the cross plots represents a pareto optimal solution. Only solutions with combined extraction greater than 2.1 $\frac{m^3}{d}$ (three times the historical rate) are shown. A) displays the spatial characteristics of each pareto-optimal artificial recharge basin. Points on B) through Z) are colored according to risk (dark blue/purple points depict risk-tolerant/low-reliability solutions; yellow points are risk-averse/high-reliability solutions). (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

encouraged to proceed in a stepwise-testing fashion to evaluate the importance or otherwise of increased rigor in the coupling strategy and representation of risk. pestpp-mou provides, for the first time, a means to systematically evaluate these coupling strategies.

The object-oriented and encapsulated design and implementation of the population generators and environmental selectors within pestpp-mou allow for additional generators and selectors to be added with relative ease. This is important as the field of multi-objective optimization is the focus of much research and it is expected that more efficient formulations will continue to be developed.

Disclaimer

Any use of trade, firm, or product names is for descriptive purposes only and does not imply endorsement by the U.S. Government.

Software availability

The software tool presented herein is available within the open-source PEST++ code suite available at https://github.com/usgs/pestpp as a C++ code named pestpp-mou; Pre-compiled, statically-linked (e.g. stand alone) binaries for Windows, Linux, and MacOS are available, as are cmake, make, and visual studio solutions. The analytical benchmark application and the density-dependent groundwater flow application input files are available in the pestpp-mou continuous integration testing repository at https://github.com/pestpp/pestpp-mou_benchmarks. We note the continuous integration testing suite for pestpp-mou includes many more benchmark problems, including stochastic variants of many of the test problems presented in (Zitzler et al., 2000) and (Deb et al., 2002). The code repository also includes a user's manual that provides additional guidance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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